Session 4.3

Cohorts Under Aggregate Risk

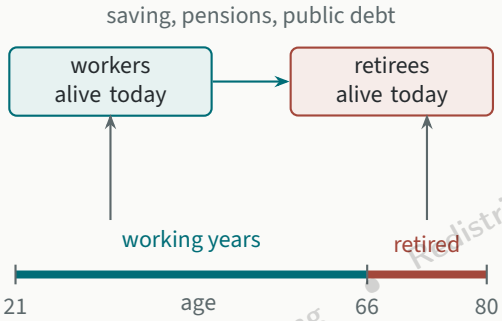
From Krusell–Smith to overlapping generations: a finite state, a network that clears a market, and why idiosyncratic risk sends us to histories



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From Krusell–Smith to Cohorts



- Deck 4.2: households live for ever; the state is $(a, \varepsilon; z, \Gamma)$ and Γ must be forecast.
- **Overlapping generations** (OLG): lives last J periods, and cohorts of every age $j = 1, \ldots, J$ coexist.
- Newborns hold nothing, the oldest leave, and Γ becomes a *stack*: one distribution Γ_j for each age.
- **Same question as Deck 4.2, what stands for Γ, with a new structure to exploit.**

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Source: Samuelson, *JPE* (1958); Diamond, *AER* (1965); Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987).

Two Kinds of Risk, Four Economies

	no aggregate risk	**aggregate risk z_t**
idiosyncratic ε	**Part A**: the classical life cycle; Γ a stack of cohort histograms; paths after one shock	**Part C**: every cohort a histogram, J of them moving with z; the state explodes
no ε	one household per cohort, no risk: a deterministic life cycle	**Part B**: one household per cohort; the state is $1 + 2J = 15$ numbers; a network

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- Parts A and B each drop one kind of risk, and each can then carry Γ exactly.
- A path answers one unanticipated shock. With z_t drawn every period the policy must read (z, Γ): Krusell and Smith's problem, now with a stack.
- **Part C puts both risks back, and borrows Deck 4.2's third method to survive it.**

Source: Krusell & Smith, *JPE* (1998); Ríos-Rull, *RES* (1996); Azinovic-Yang & Žemličková (2025).

Finite Lives: The Last Period Is Known



each arrow is one endogenous-grid step, using next year's policy

- The lab's life-cycle economy: $J = 60$ cohorts, ages 21 to 80, idiosyncratic ε, a limit $a' \geq 0$, a pension b.
- The policy depends on age: $a' = g_j(a, \varepsilon)$. At the last age consumption is cash on hand.
- Given c_{j+1}, Deck 4.1's Euler step delivers c_j: 59 steps in all, from age 80 back to 21. Deck 4.1's infinitely lived household instead iterates to a fixed point: 356 sweeps from a cold start (the Session 4 lab, block A).
- **Backward induction replaces iteration to convergence: the household problem has no fixed point left.**

Source: Auerbach & Kotlikoff (1987); Carroll, *Economics Letters* (2006); the Session 4 lab, block C2. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

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Cohorts Are the Distribution



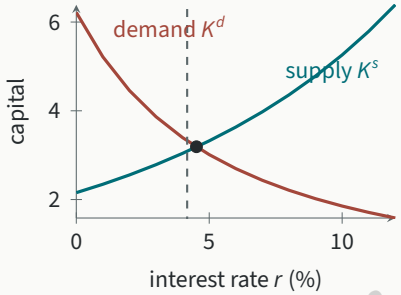
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- Γ_1 is known: newborns hold nothing, with ε drawn from its invariant distribution.
- $\Gamma_{j+1} = \Gamma_j \Phi_{g_j}$: Deck 4.1's lottery matrix, applied once for each age.
- Given prices, no iteration to an invariant distribution: 59 pushes complete the cross-section. The price is still unknown: next page.
- **The economy's distribution is the stack of the cohorts'. Part C will need exactly this stack, moving with z.**

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Source: Young, *JEDC* (2010); Ríos-Rull, *Review of Economic Studies* (1996).

The Stationary Equilibrium: One Price, Found by Bisection



The stationary equilibrium

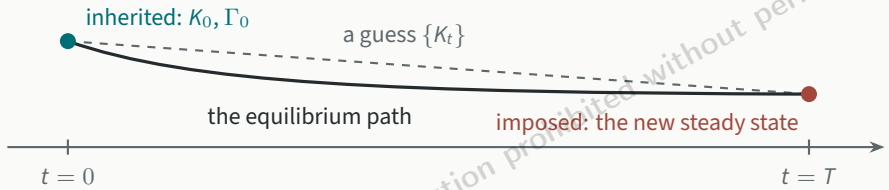
1. bracket $r \in [-7\%, 30\%]$; check the sign change
2. trial r: firm gives K/L, w; tax pays pension
3. households backward: 59 EGM steps
4. cohorts forward: 59 pushes give $K^s(r)$
5. compare with $K^d(r)$; halve the bracket
6. stop after 29 halvings; residual 6×10^{-9}

- $r^* = 4.52\%$ lies above $1/\beta - 1 = 4.17\%$ (dashed): with finite lives that bound fails; the bracket must be wide.
- **One price, one fixed point; inside it, one backward and one forward pass.** Code: the Session 4 lab, block C2.

Source: The Session 4 lab, block C2: 60 cohorts, a 40% pay-as-you-go pension.

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The Unknown Is a Path of Prices

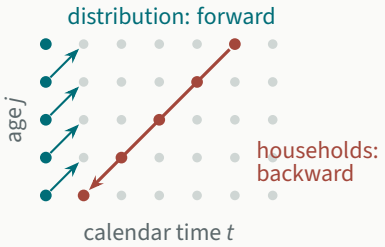


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- An unanticipated, permanent change at $t = 0$: an "MIT shock". Afterwards the path is known.
- Unknown: $\{K_t\}_{t=0}^{T}$. Prices $\{r_t, w_t\}$ follow from the firm's conditions.
- Boundary conditions at both ends: Γ_0 is inherited, and behaviour at T is the new steady state's.
- **Decisions look forward and distributions look back. No single pass can do both.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger, *RED* (1999).

Update, Damp, Repeat



The transition loop

1. solve both steady states; choose the horizon T
2. guess $\{K_t\}$; prices follow from it
3. households backward along each cohort's diagonal
4. distribution forward from Γ_0: the supply K_t^s
5. stop when $\max_t |K_t^s - K_t|/K_t$ is small
6. else $K_t \leftarrow (1-d)K_t + dK_t^s$, K_0 fixed

- Damping $d = 0.5$: a higher guess lowers r_t and the saving meant to support it, so a full step can oscillate.
- the Session 4 lab, block C runs this loop for Deck 4.1's infinitely lived economy after a 20% tax on capital income: 91 passes to 1×10^{-7}, $T = 300$. **The outer loop is a fixed point in a sequence.**

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Source: Auerbach & Kotlikoff (1987); Conesa & Krueger (1999); the Session 4 lab, block C. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

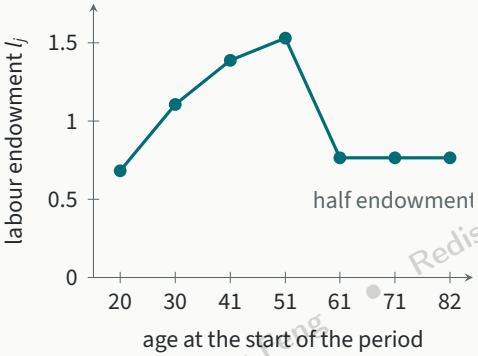
Measuring Who Gains

- **Consumption-equivalent variation** $g(a, \varepsilon)$: scale the old consumption plan by $(1+g)$ at every date and in every state, until its value matches the new one. $g > 0$ is a gain.
- With CRRA utility V scales by $(1+g)^{1-\sigma}$, so $g(a, \varepsilon) = \left[V_1(a, \varepsilon)/V_0(a, \varepsilon)\right]^{1/(1-\sigma)} - 1$.
- Everything turns on *which* V_1. Along the path: the value at $t = 0$ of living through the transition. Between steady states: the value of waking up in the new one.
- Average with the distribution of the people who are asked: Γ_0, the households alive at the reform.
- **One formula, two numbers. The difference is whether the path is counted.**

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Source: Lucas, *Models of Business Cycles* (1987); Domeij & Heathcote, *International Economic Review* (2004).

The Economy: Seven Cohorts, Two Assets

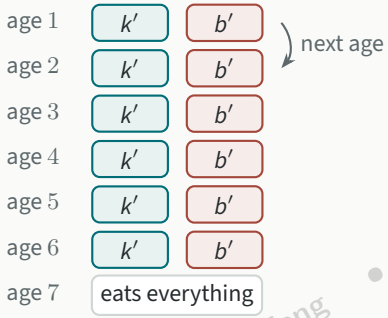


- $J = 7$ cohorts; one period is $72/J \approx 10.3$ years, ages 20 to 82. Endowments l_j (left), summing to J.
- CRRA $\gamma = 2$; $\beta = 0.96^{72/J} \approx 0.66$. Two assets: capital k and a one-period bond b.
- TFP: $\log z' = \rho \log z + \sigma\epsilon$, $\rho \approx 0.19$, $\sigma \approx 0.056$: annual 0.85 and 0.03, rescaled.
- Depreciation moves with TFP: $\delta_t = \min\{\delta \frac{2}{1+z_t}, 1\}$, $\delta \approx 0.66$ a period (10% a year). **A bad draw also destroys capital.**

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Source: Part B follows Azinovic-Yang & Žemlička, "Economics-Inspired Neural Networks with Stabilizing Homotopies", arXiv 2303.14802, and its OLG notebook (the calibration cells). In PyTorch, scaled for a CPU: the Session 4 lab, block G.

The Household: Capital, Bonds, Adjustment Costs



- Age j with holdings (k, b) chooses (k', b'):

$$c + p\,b' + k' + \psi_k (k' - k)^2 = l_j w + b + (1 - \delta + r)\,k.$$

- $k' \geq 0$; $b' \geq b_{\min} = -0.05$; the last cohort saves nothing. $\psi_k = 0.1$: changing one's *own* capital is costly.
- $J - 1 = 6$ ages choose: $2(J-1) = 12$ saving decisions.
- **No ε: everyone of an age is identical, so a cohort is one household.**

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Source: Notebook markdown "Households" and the budget in *E*; *psi_k*, *b_min_target*.

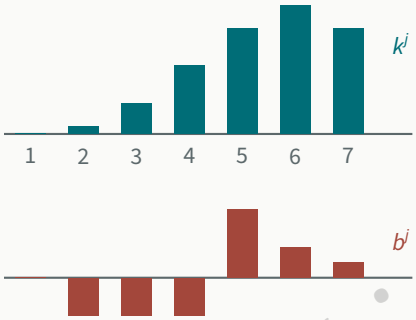
Firm and Markets

- Firm: $Y = z\,K^{\alpha}L^{1-\alpha}$, $\alpha = 1/3$; $r = \alpha Y/K$, $w = (1-\alpha)Y/L$.
- Aggregates: $K = \sum_j k^j$, $L = \sum_j l_j = J = 7$, fixed.
- Bonds in zero net supply: $\sum_j b'^j = 0$. The bond price p is the one price not given by a marginal product.
- Goods clear by Walras' law.
- The notebook's text swaps the formulas for r and w; the code, *get_firm_side*, is as written here.
- **Two assets, one of them priced by the market: a network must now return a price as well as policies.**

Source: Notebook *get_firm_side*; market-clearing markdown.

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The State Is Fifteen Numbers



age j; bars schematic

- The distribution is a vector, $\Gamma = (k^1, \ldots, k^J, b^1, \ldots, b^J) \in \mathbb{R}^{2J}$; the input $x = (z, \Gamma)$ has $d = 1 + 2J = 15$ numbers.
- Unknowns: $J - 1$ capital and $J - 1$ bond policies and the price, as functions of x: $n_{\text{out}} = 13$.
- Equilibrium: $2(J - 1) = 12$ KKT conditions (six saving ages, two assets) and bond clearing, at every x.
- **Unlike Deck 4.2: Γ is exact. No moment, no histogram, no history.**

Source: Notebook “State Variable and Equilibrium Functions”; the index constants; the code keeps $k^1 = b^1 = 0$ as inputs.

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Tomorrow's Distribution Is a Shift



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- Age j's choice today is age $j+1$'s holding tomorrow; a newborn enters with $(0, 0)$; age J leaves. So $\Gamma' = (0, k'^{1}, \ldots, k'^{J-1}, 0, b'^{1}, \ldots, b'^{J-1})$: $2J$ numbers again.
- Γ' is known once today's choices are made: the only randomness in $x' = (z', \Gamma')$ is z', one successor per quadrature node.
- **Unlike Deck 4.2's lottery: no mass to split between grid points. The policy's output *is* the next state.**

Source: Notebook *ForwardStep* and the "next period wake-up assets" in *E*.

Equilibrium Conditions as Fischer–Burmeister Residuals

- KKT conditions for age $j < J$, written as relative consumption errors, u'^{-1} the inverse marginal utility:

$$\varepsilon_k = \psi^{FB}\Big(\frac{u'^{-1}\big(\beta\,\mathbb{E}[u'(c')\,(1-\delta'+r'+2\psi_k(k''-k'))]\,/\,(1+2\psi_k(k'-k))\big)}{c} - 1,\ \frac{k'}{c}\Big),$$

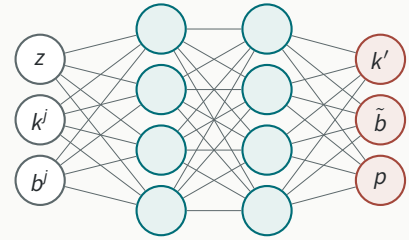
$$\varepsilon_b = \psi^{FB}\Big(\frac{u'^{-1}\big(\beta\,\mathbb{E}[u'(c')]\,/\,p\big)}{c} - 1,\ \frac{b'-b_{\min}}{c}\Big), \qquad \psi^{FB}(a,b) = a + b - \sqrt{a^2+b^2}.$$

- $\psi^{FB} = 0$ exactly when $a \geq 0$, $b \geq 0$, $ab = 0$: the Euler equation holds, or the limit binds.
- $\mathbb{E}$: $n_{GH} = 8$ Gauss–Hermite nodes over z'. c' and k'' from the same network at x'.
- **As Deck 4.2's Deep Equilibrium Nets, one period ahead; now two assets and $2(J-1) = 12$ residuals per state.**

Source: Notebook *E* (capital and bond terms), *fischer_burmeister*, *num_GH_tfp*; Maliar, Maliar & Winant, *JME* (2021).

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The Network: Fifteen Numbers In, Thirteen Out

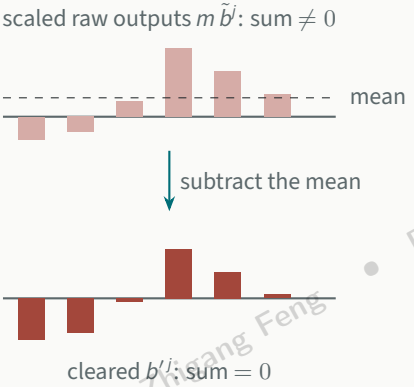


- One network for all $n_{\text{out}} = 13$ functions: two shared selu layers of $n_{\text{hid}} = 10J = 70$ units, so ages share what they learn.
- Capital: $k' = 0.1\,\text{softplus}(o_k) \geq 0$, so no short sales by construction.
- Price: $p = \text{softplus}(o_p) > 0$.
- Bonds: the raw $\tilde{b}$ goes through a clearing layer (next page).
- **Consumption is not an output: it is backed out of the budget.**

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Source: Notebook *base_net_construct*, *S_fun*; hidden width $N = 10\times N_cohort$.

A Layer That Clears the Bond Market



- From the $J-1=6$ raw outputs $\tilde{b}^j$:

$$b'^j = m\,\tilde{b}^j - \frac{1}{J-1}\sum_i m\,\tilde{b}^i.$$

- Each age keeps its scaled output minus the six-age average. Summed over j, the terms cancel: $\sum_j b'^j = m\sum_j \tilde{b}^j - m\sum_i \tilde{b}^i = 0$.
- m: set by hand, not learned (next page).
- **Unlike Deck 4.2: the market clears by construction, not by a loss term.**

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Source: Notebook *S_fun*: the mean excess over the net supply (zero here) is subtracted; the paper (market-clearing layers).

The Mask and the Price: One Set by Hand, One Pinned by the Loss

stages	m	**bond positions** $b'^j = m\,(\tilde{b}^j - \bar{\tilde{b}})$, $\bar{\tilde{b}}$ the six-age average
1–2	0	all zero, whatever the network outputs: no bonds yet
3 to the end	0.1	the raw outputs shrunk tenfold: bonds enter small

- **Mask** m: not an output, not trained. One of five numbers set by hand at each stage of training (the homotopy, later): bond supply 0, the limit $b_{\min}$, m, and two loss weights.
- Why shrink: freshly switched on, the bond outputs are untrained; scaled by 0.1 they barely move the economy just solved. Afterwards 0.1 is a fixed scale.
- **Who sets** p**?** Not clearing, which now holds at any p. The $J - 1 = 6$ bond Euler residuals do: six conditions for six unknowns, p and the positions, which have five free directions once they sum to zero.
- **A price is an output pinned down by the loss. With no bonds traded, nothing pins it.**

Source: Notebook *S_fun*, the *Homotopy* array, the stage cells; a comment says 0.01, the code sets 0.1.

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One Call at One State: Every Cohort's Choice, Then Every Budget

age j	holds today	chosen by the network	consumes, from its budget
1	$(0, 0)$: newborn	k'^1, b'^1	c^1
$2, \ldots, 6$	(k^j, b^j)	k'^j, b'^j	c^j
7	(k^7, b^7)	nothing: $k'^7 = b'^7 = 0$	c^7: all that is left
all ages	z; then K, r, w	one price p	

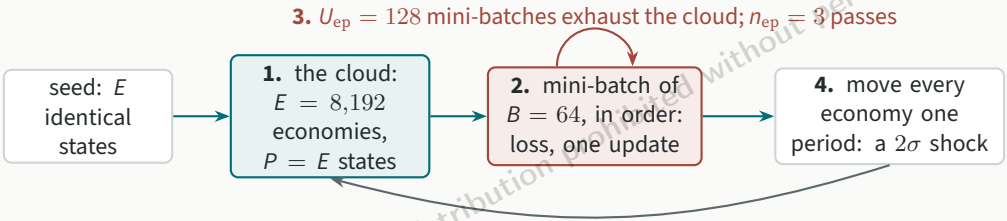
$$c^j = l_j w + b^j + (1 - \delta + r)\, k^j - p\, b'^j - k'^j - \psi_k \left(k'^j - k^j\right)^2, \qquad j = 1, \ldots, J.$$

- One call at one state $x = (z, \Gamma)$ returns all $n_{\text{out}} = 13$ numbers: every saving age's k'^j, b'^j and p. A cohort is one household, so this is every household's choice.
- No maximisation, no EGM: k'^j is the network's guess; the Euler residuals judge it.
- **Unlike Deck 4.2's Method 2, whose network maps one household's $(\varepsilon, a; z, \Gamma)$ to its (a', λ): here one call is the whole economy.**

Source: Notebook *S_fun* (the 13 outputs) and the budget in *E*; as in the code, $k^1 = b^1 = 0$ and the last cohort saves nothing.

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The Training Loop in One Picture

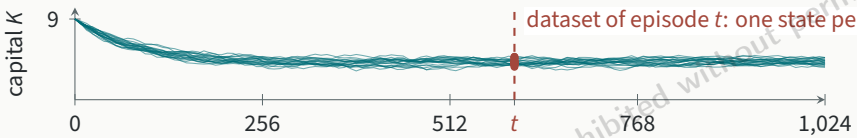


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- **1. Samples**: $E = 8{,}192$ economies side by side; their current states are the dataset.
- **2. Mini-batch**: the next $B = 64$ states in order, the loss at them, one gradient step.
- **3. Epoch**: $U_{\text{ep}} = E/B = 128$ mini-batches use the cloud once; $n_{\text{ep}} = 3$ passes.
- **4. Episode**: every economy moves one period; stage 1 alone runs $n_1 = 1{,}024$ episodes.

Source: Notebook *CloudMethod*, *Episode*, *TrainingLoop*; *n_cloud*, *n_minib*, *epochs_epis*. Symbols: the symbol page.

1. Samples: 8,192 Economies From One Seed, Never Restarted



- Each line is one economy's capital; 20 of the $E = 8{,}192$ are drawn. All start identical: $\log z = 0$, $k^2, \ldots, k^7 = 1.5$ ($K = 9$), no bonds; they differ only by their shocks.
- Dataset of episode t: the slice, $P = E$ states, one per economy, each $x = (z, \Gamma)$.
- Train, then move: $U_{\text{epis}} = 384$ updates on the slice, then every economy moves one period with the updated network. The seed is used once; no run restarts.
- **While the network trains on slice t, only that slice exists: earlier ones are discarded, later ones not yet simulated.**

Source: Notebook settings ($n_cloud = 2^{13}$), the seed (*representative_input*), *Episode*, *CloudMethod*. Paths schematic.

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2–3. Mini-Batches in Order, the Cloud Exhausted, Three Passes

One episode's cloud, then $U_{\text{epis}} = 384$ updates

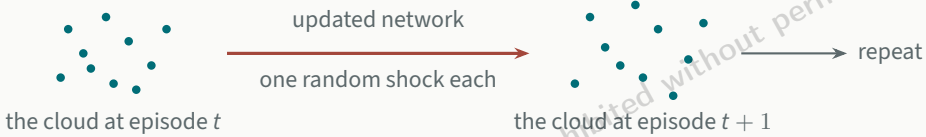
1. the cloud: $P = E = 8{,}192$ states $x = (z, \Gamma)$, one per economy, in the economies' order
2. take the next $B = 64$ states in order: economies 1–64, then 65–128, and so on
3. one Adam update of θ on the loss of the next pages, rate 10^{-5}
4. repeat 2–3; after $U_{\text{ep}} = E/B = 128$ updates every state is used once: *exhausted*
5. twice more, the same mini-batches in the same order: $n_{\text{ep}} = 3$ passes, $U_{\text{epis}} = n_{\text{ep}} U_{\text{ep}} = 384$
6. then the cloud moves one period (next page); the old states are discarded

- **Epoch**: one pass over the cloud, U_{ep} updates. Each state enters the loss $n_{\text{ep}} = 3$ times, never shuffled; Deck 4.2's Method 2 used each point once, DeepHAM's critic reused its rows 201 times.
- **One Adam update of θ per mini-batch:** $U_{\text{epis}} = n_{\text{ep}} E/B = 384$ **an episode;** $n_1 U_{\text{epis}} = 1{,}024 \times 384 = 393{,}216$ **in stage 1.**

Source: Notebook *TrainingLoop* ($n_minib = 2^6$, $epochs_epis = 3$; the cloud repeated and reshaped in order), *Update*.

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4. Episodes: the Cloud Moves One Period



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- After the $U_{\text{epis}} = 384$ updates, each economy moves one period: its policy from the updated network, its own shock drawn at $m_\sigma = 2$ times σ, truncated at ± 3 s.d. (the loss uses σ, so the data reach past the economy's usual range).
- The moved states replace the old ones and are the next episode's dataset.
- Stage 1, the capital-only economy, runs $n_1 = 1{,}024$ episodes; the later stages continue from the same E economies.
- **This is the only simulation step. The look-ahead inside the loss (next page) computes an expectation and moves nothing.**

Source: Notebook *Episode* (*truncated_normal*, *shock_mult* $= 2$), *ForwardStep*, *CloudMethod*; Bretscher et al. (2022).

5. Inside One Update: One Mini-Batch, Line by Line

One update on a mini-batch of $B = 64$ states

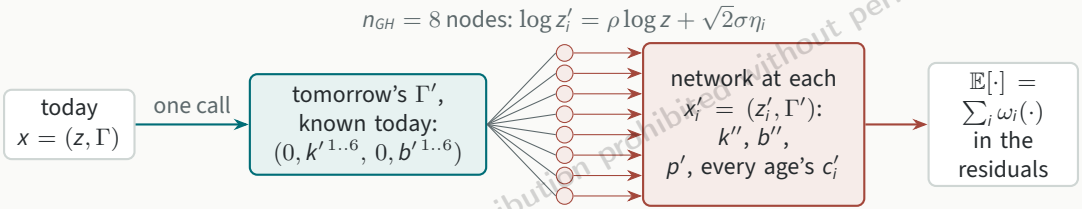
1 take the next $B = 64$ states $x = (z, \Gamma)$ of the cloud, in order
2 for each state, one call gives every saving age's k'^j, b'^j and p; each c^j from its budget
3 the same economy tomorrow at $n_{GH} = 8$ TFP nodes: $x'_i = (\rho \log z + \sqrt{2}\sigma\eta_i,\ \Gamma'), \Gamma'$ shared
4 call the network at each x'_i: k'', b'', p'; then every age's c'_i from the budget at $t + 1$
5 $\mathbb{E}[\cdot] = \sum_i \omega_i(\cdot)$; the residuals ε_k, ε_b for ages $1, \ldots, J - 1$
6 mean loss over the B; gradient through all $1 + n_{GH} = 9$ calls, k', b' inside x'_i too; Adam, 10^{-5}

- Steps 3–5 are the look-ahead (next two pages): eight successors of one economy, not households, not simulated. Per state $1 + n_{GH} = 9$ calls; per update $n_{\text{call}} = (1 + n_{GH})B = 576$.
- **Unlike Deck 4.2's Method 2, whose successors are stored data: here they are rebuilt inside the loss, so the gradient sees the move to x'.**

Source: Notebook *E*, *vE*, *Loss*, *GradAndLoss*, *Update*; *num_GH_tfp* $= 8$.

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5. The Look-Ahead: One Economy, Eight Successors

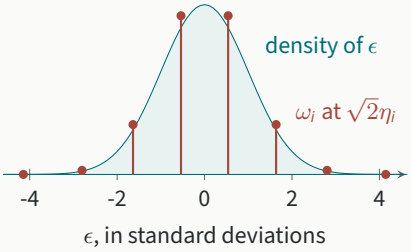


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- A successor is tomorrow's state of this economy at one TFP node: $x_i' = (\log z_i', \Gamma')$, $i = 1, \ldots, n_{GH}$; only z' differs across the eight.
- $\Gamma' = (0, k'^1, \ldots, k'^6, 0, b'^1, \ldots, b'^6)$ in all eight: newborns hold nothing, age j's choice is age $j+1$'s holding, age 7 has left.
- **Not households, unlike Deck 4.2's $n_{\text{pt}} = 8$ evaluation points, and not simulated: they serve the expectation (next page), then are discarded.**

Source: Notebook *E*: *x_next* stacks the eight successors in an 8×15 array whose rows differ only in the first entry.

5. The Look-Ahead: Eight Gauss–Hermite Nodes



$$\mathbb{E}[g(\epsilon)] \approx \sum_{i=1}^{n_{GH}} \omega_i \, g\left(\sqrt{2}\,\eta_i\right)$$

- The shock: $\log z' = \rho \log z + \sigma\epsilon$, $\epsilon \sim N(0,1)$.
- η_i: roots of the Hermite polynomial H_8; ω_i: its weights over $\sqrt{\pi}$, summing to 1.
- $\epsilon = \sqrt{2}\eta$ turns the normal density into the rule's weight $e^{-\eta^2}$: hence the $\sqrt{2}$.
- Exact for polynomials of degree $\leq 2n_{GH} - 1 = 15$.
- In the residuals: $\mathbb{E}[u'(c'^{j+1})] = \sum_i \omega_i \, u'(c_i'^{j+1})$.
- **Fixed nodes: no sampling noise.**

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Source: Notebook *hermgauss* (NumPy) for *eta_tfp*, *GH_weights_tfp*; the plotted nodes and weights are computed.

5. Inside One Update: the Loss

- A mini-batch $\mathcal{B}$ is $B = 64$ states x (a setting, 2^6); residuals are averaged over the B states and the $J - 1$ ages:

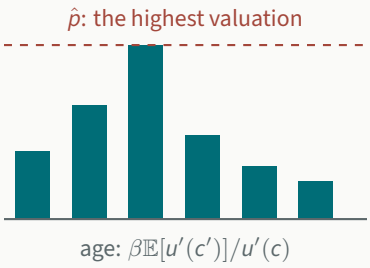
$$\mathcal{L}(\theta; \mathcal{B}) = \tfrac{1}{2}\,\overline{\varepsilon_k^2} + \tfrac{1}{2}\,w_b\,\overline{\varepsilon_b^2} + w_{\text{pre}}\,\overline{\varepsilon_{\text{pre}}^2} + 100\,\overline{\varepsilon_c^2}.$$

- $\varepsilon_c = \max(c, 10^{-4}) - c$: a network that implies negative consumption is pushed back hard.
- $\varepsilon_{\text{pre}} = p - \text{sg}(\hat{p})$: a target price, with its gradient stopped.
- w_b, w_{pre} and the mask m are switched by the stage. One Adam step per mini-batch: $U = 3{,}970{,}560$ in all.
- **Unlike Deck 4.2's methods, the loss weights change during training: the loss is a path of problems.**

Source: Notebook *Loss*, *GradAndLoss*, *E* (*eps_cnn*, *eps_bpr*).

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Why a Homotopy: a Bond Nobody Can Trade Has No Price

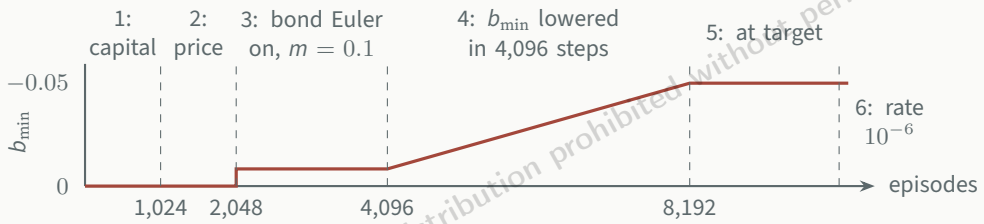


- At $b_{\min} = 0$ and zero net supply no one can hold a bond, so any high enough price clears the market: p is not pinned down.
- Its natural limit: the **zero-liquidity price** $\hat{p} = \max_j \beta\,\mathbb{E}[u'(c'^{j+1})]/u'(c^j)$, what the most eager saver would pay for the first bond.
- Then relax $b_{\min}$ a little at a time; each problem starts from the last one's solution.
- **Solving the full model from a random start is where state-space training tends to fail.**

Source: Notebook "Pretrain bond price", *p_b_implied*; the paper.

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The Homotopy in Six Stages



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- 1: capital only, $w_b = w_{\text{pre}} = m = 0$. 2: $w_{\text{pre}} = 1$, learn $\hat{p}$. 3: $w_b = 1$, $w_{\text{pre}} = 0$, $m = 0.1$, $b_{\min} = -0.05/(J-1)$.
- 4: $b_{\min}$ lowered to -0.05, one episode a step. 5: at the target. 6: the rate cut to a tenth.
- **Each stage inherits network, cloud and optimiser: the same $E = 8{,}192$ economies run through all six. Episodes and updates: next page.**

Source: Notebook, “Initialize Capital-Only Model” to fine-tuning; a comment says mask 0.01, the code sets 0.1.

The Six Stages in Episodes and Updates

stage s	what is switched	n_s	$U_{\text{epis}}\, n_s$	rate
1 capital only	$w_b = w_{\text{pre}} = m = 0$; from the seed	1,024	393,216	10^{-5}
2 learn the price	$w_{\text{pre}} = 1$: p fitted to $\hat{p}$	1,024	393,216	10^{-5}
3 bonds on	$w_b = 1, w_{\text{pre}} = 0, m = 0.1, b_{\min} = -0.05/6$	2,048	786,432	10^{-5}
4 lower the limit	$b_{\min}$ down to -0.05, one episode a step	4,096	1,572,864	10^{-5}
5 at the target	nothing: train on at $b_{\min} = -0.05$	2,024	777,216	10^{-5}
6 fine-tune	the rate only	124	47,616	10^{-6}
all	$n_{\text{epis}} = \sum_s n_s$; $U = n_{\text{epis}} U_{\text{epis}}$	10,340	3,970,560	

- $U_{\text{epis}} = 384$ updates an episode throughout, so stage s makes $U_{\text{epis}} n_s$ updates: $U = n_{\text{epis}} U_{\text{epis}} = 10{,}340 \times 384 = 3{,}970{,}560$ in all.
- **The notebook times each stage in minutes on one T4 GPU, in comments; nothing else about the run is stored.**

Source: Notebook *cloud_steps_cap*, *cloud_steps_first*, *homotopy_steps*, the final and fine-tuning calls; *Gamma*, *Gamma_fine*.

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Three Clocks, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
network	each mini-batch of	nothing simulated: $n_{GH} = 8$ successors inside the loss,
update	$B = 64$, in order	discarded; one Adam step on θ
episode	$U_{\text{epis}} = 384$ updates	one period of all E from their own states, network frozen, 2σ shocks; the moved cloud replaces the old
stage	after its n_s episodes	nothing: $b_{\min}$, m, w_b, w_{pre} or the rate switch; network, cloud and optimiser carry over

The algorithm: $U = 3{,}970{,}560$ updates

1 seed $E = 8{,}192$ identical states and a random network
2 for each stage s, set $(b_{\min}, m, w_b, w_{\text{pre}}, \text{rate})$; for each of its n_s episodes:
3 $\quad U_{\text{epis}} = 384$ updates: $n_{\text{ep}} = 3$ passes of $U_{\text{ep}} = 128$ mini-batches of $B = 64$ states
4 $\quad$ move the cloud one period with fresh shocks at 2σ

Source: Notebook *CloudMethod*, *Episode*, *TrainingLoop*; the stage cells.

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The Symbols

symbol	value	what it counts, and the formula
J	7	cohorts (*N_cohort*); one period is $72/J \approx 10.3$ years
$\beta, \delta, \rho, \sigma$	0.66, 0.66, 0.19, 0.056	$0.96^{72/J}$; $1-0.9^{72/J}$; $0.85^{72/J}$; $0.03\sqrt{\sum_{i<10} 0.85^{2i}}$
d, n_{out}	15, 13	inputs (z, Γ), $1+2J$; outputs, $2(J-1)+1$
n_{hid}, n_{GH}	70, 8	selu units per hidden layer, $10J$; quadrature nodes for z'
E, P	8,192, 8,192	economies side by side, 2^{13}; the dataset, $P = E$
B, U_{ep}	64, 128	mini-batch, 2^6, in order; updates that use the cloud once, E/B
n_{ep}, U_{epis}	3, 384	passes over the cloud, same order; updates an episode, $n_{\text{ep}} U_{\text{ep}}$
n_{call}, m_σ	576, 2	network calls per update, $(1+n_{GH})B$; shock multiple
n_{epis}	10,340	episodes in all; stage s runs n_s: $\sum_s n_s = 1{,}024 + 1{,}024 + 2{,}048 + 4{,}096 + 2{,}024 + 124$
U	3,970,560	updates in all, $n_{\text{epis}} U_{\text{epis}}$
m, $b_{\min}$	0.1, -0.05	mask on bonds; the limit, from $-0.05/(J-1)$ in 4,096 steps
rate	$10^{-5}, 10^{-6}$	Adam; in the last 124 episodes

Source: Notebook settings *N_cohort*, *n_cloud*, *n_minib*, *epochs_epis*, *num_GH_tfp*, *shock_mult*, the stage cells, *Gamma*.

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Four Methods, One Table

	DeepHAM	Deep Eq. Nets, KS	Sequence space	Deep Eq. Nets, OLG (here)
what stands for Γ	learned moments of $N = 50$	two $N_b = 100$-point histograms	the last $T_{\text{hist}} = 50$ shocks	Γ **itself:** $2J = 14$ **numbers**
what is minimised	minus lifetime utility; a value regression	Euler and KKT residuals	an FB Euler residual	FB residuals; bond clearing built in
where states come from	$E = 384$ panels, refreshed	$E = 16$ tracks, carried forward	a pool of $P_{\max} = 128$	a cloud of $E = 8{,}192$, moved one period
what the gradient passes through	$T_{\text{unroll}} = 150$ periods, simulated	one period, at stored states	one period; prices fixed	one period, through the computed x'

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- **Row 1 is why the OLG is easy here: with no ε, nothing has to stand for Γ.**

Source: Deck 4.2, *Four Questions Every Deep Method Answers* and *Three Pipelines, the Same Letters*; the homotopy notebook.

What Changed From Deck 4.2

- **An exact state.** Cohorts without ε are points, so the distribution is a vector of $2J = 14$ numbers, and its law of motion is a shift.
- **A price as an output.** Two assets need a bond price; the network returns it, and a layer clears the market exactly.
- **A path of problems.** Capital first, then a price, then bonds that trade a little more each step.
- **Wide and shallow simulation.** $E = 8{,}192$ economies moved one period at a time, at twice the shock's s.d.
- **Successors differentiated.** The loss builds x' from the policy; the gradient sees the move.
- **All five rest on the first. Take it away and the rest has nothing to stand on.**

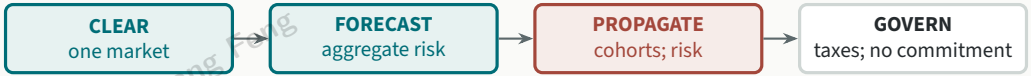
Source: The homotopy notebook; Deck 4.2.

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Session 4.3

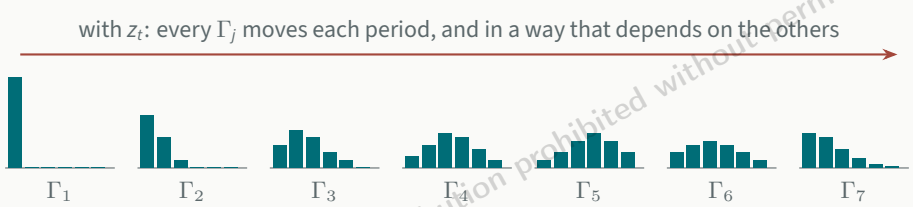
Put Idiosyncratic Risk Back

After Azinovic-Yang & Žemlička, *Deep Learning in the Sequence Space*: every cohort becomes a distribution, and the state explodes



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Each Cohort Becomes a Histogram

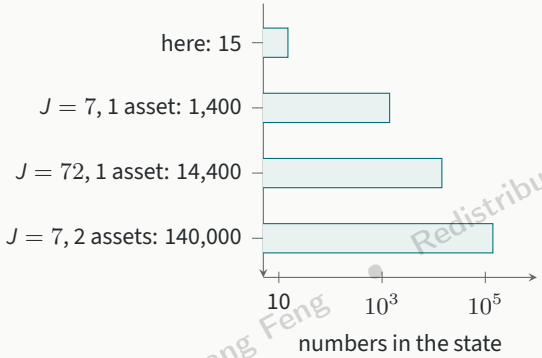


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- With ε, two households of the same age differ. A cohort is a distribution $\Gamma_j(\varepsilon, k)$ again, as in Part A.
- The aggregate state: $(z, \Gamma_1, \ldots, \Gamma_J)$. Part A's stack, but no longer stationary: it moves with z.
- **Part A had the stack without z; Part B had z without the stack. Now both.**

Source: Part C follows Azinovic-Yang & Žemlička, "Deep Learning in the Sequence Space", working paper, arXiv 2509.13623 (2025), §6, with their slides (2026); here p. 113. In PyTorch, scaled for a CPU: the Session 4 lab, block H.

Count the Inputs



- Two income states, $N_b = 100$ asset points per cohort: $J \times 2 \times N_b = 7 \times 2 \times 100 = 1{,}400$.
- With the published OLG's $J = 72$ cohorts: $72 \times 2 \times 100 = 14{,}400$.
- Two assets on a 100×100 grid: $7 \times 2 \times 10^4 = 140{,}000$.
- **A state-space network reads all of it, and its training states move whenever the policy does.**

Source: Arithmetic. Cohort count: their slides, p. 13.

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Why Deck 4.2's First Two Answers Strain

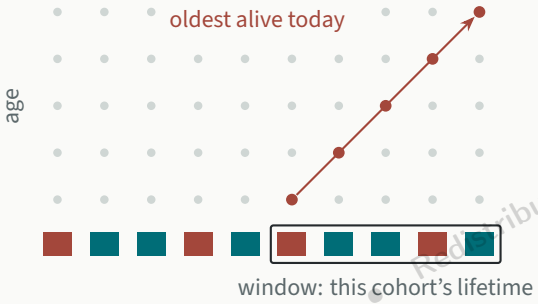
	Krusell–Smith, one distribution	**OLG, J distributions**
classical rule	forecast K' from K	forecast every cohort's moments, or prices from many of them
DeepHAM	one panel of $N = 50$, one learned moment	a panel for each age; moments per age
Deep Eq. Nets	$2N_b = 200$ masses as input	$J \times 2N_b$ masses, drifting with the policy
sequence space	the last $T_{\text{hist}} = 50$ shocks	**the same window of shocks**

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- The first three grow with the number of ages J; the last does not, because its input never contained Γ.
- Deck 4.2, *What the Exogenous Domain Buys*: large inputs that shift with the policy feed the unstable loop.
- **Only the history keeps its size when cohorts are added.**

Source: Deck 4.2; their slides, pp. 11, 23–24.

The Way Out: the History Already Holds Every Cohort

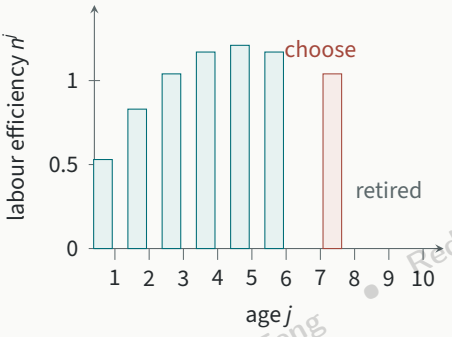


- Everyone alive was born within the last J periods: Γ_j is built by the shocks of its lifetime, from its starting wealth.
- Older shocks act only through prices and inheritances (bequests, in the model ahead): Deck 4.2's **memory loss**.
- So the stack is, approximately, a function of recent shocks: a window at least a lifetime long, $T_{\text{hist}} \geq J$.
- **Published sweep, 1 to 256 lags: accuracy improves with the window.**

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Source: Intuition ours; evidence: their slides, pp. 7–8, 20–21, 91 (the paper's 72-cohort OLG).

The Economy: Ten Cohorts, Income Risk, a Retirement Choice

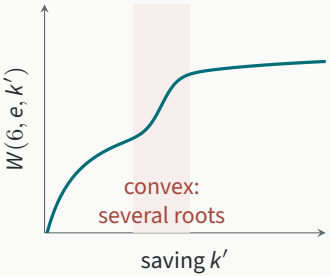


- $J = 10$ cohorts of six years: ages 1–6 work, age 7 chooses to work or retire (logit, scale 0.08), 8–10 are retired; bequests become newborns' assets.
- Income $e \in \{0.5, 1, 1.5\}$, $N_e = 3$, Markov; one asset, capital, $k \geq 0$ on $N_k = 300$ points.
- Aggregate risk: $n_z = 3$ regimes; the third is a disaster, depreciation 0.397 instead of 0.265.
- State (z, Γ), $\Gamma(j, e, k)$: $JN_eN_k = 9{,}000$ masses. Input: the last $T_{\text{hist}} = 100$ regimes.
- **The network never reads Γ.**

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Source: The paper, §6.1–6.2; their slides, pp. 105–113. The paper numbers ages 0–9; its implementation text says 200 asset points, its parameter table 300.

Why the Network Outputs Continuation Values

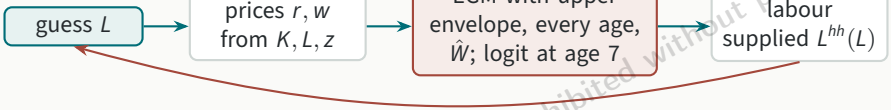


- Age 6 looks ahead to retirement, so its value of saving is convex in places (left): several Euler roots; a residual can pick the wrong one.
- So the network outputs **continuation values** $\hat{W}(j, e, k') \approx \mathbb{E}[V(j{+}1, e', k')]$ on the grid, ages $1, \ldots, J-1$; EGM with an **upper envelope** keeps the best root.
- Increasing by construction: a first value plus softplus increments. One call gives all $(J-1)N_eN_k = 8{,}100$ values: operator learning.
- **Unlike Deck 4.2's tutorial: no consumption output, no residual.**

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Source: The paper, §6.3; their slides, pp. 110, 114–120; Fella (2014); Druedahl & Jørgensen (2017).

Within the Period: Policies, and a Labour Market That Must Clear



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- K is fixed by Γ; L is not: whether age 7 works depends on the wage, and the wage on L.
- $n_{\text{Newton}} = 5$ steps, each an EGM solve of all J cohorts, ED' by automatic differentiation; start $L^{\text{work}} + \hat{\lambda}\, L^{7,\max}$, $\hat{\lambda}$ the network's guess of age 7's share.
- Result: prices, every age's saving rule $g^j(e,k)$, the work probabilities $p^{\text{work}}(e,k)$, and the realised share λ.
- **Unlike Part B, where a layer cleared the bond market exactly: here a price is found by iteration inside every evaluation.**

Source: The paper, §6.3 (Step 0 of the forward simulation) and its hyperparameters.

Moving the Distribution One Period

- Young's lottery with income transitions, for a saving rule g and masses $m(i,j)$ at (e_i, k_j):

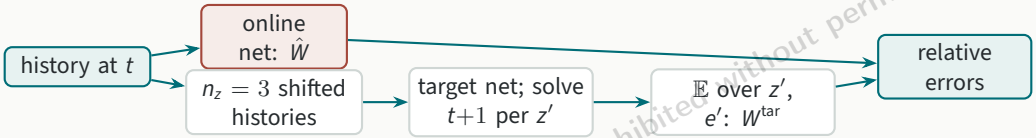
$$[\mathcal{T}(g,m)](i',j') = \sum_{i,j} m(i,j)\,\Pi^{e}_{ii'}\big(\omega^{-}\,\mathbf{1}\{j'=j^{-}\} + \omega^{+}\,\mathbf{1}\{j'=j^{+}\}\big),$$

  $j^{\pm}$ the grid points around $g(e_i,k_j)$, $\omega^{\pm}$ the lottery weights of Deck 4.2.
- Aging: $\Gamma'_{j+1} = \mathcal{T}(g^{j},\Gamma_j)$. Age 7 splits: $\mathcal{T}(g^{\text{work}}, p^{\text{work}} \odot \Gamma_7) + \mathcal{T}(g^{\text{ret}}, (1-p^{\text{work}}) \odot \Gamma_7)$.
- Newborns: $\Gamma'_1 = \mathcal{T}(g^{J},\Gamma_J)$, the bequests.
- Γ' does not depend on z': $n_z = 3$ candidate successors (z',Γ'), each with its history shifted by z'.
- **The same lottery as Deck 4.2's Methods 2 and 3, applied age by age.**

Source: The paper, §6.3 (Steps 1–3 of the forward simulation); Young, *JEDC* (2010).

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The Loss: Targets From a Slower Copy of the Network

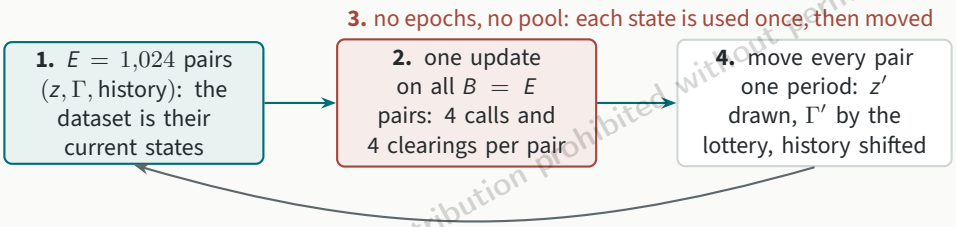


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- Target: $W^{\text{tar}}(j, e, k') = \sum_{z'} \Pi^{z}_{zz'} \sum_{e'} \Pi^{e}_{ee'} V_{t+1}(j{+}1, e', k' \mid z'; \bar{\rho})$, $t{+}1$ solved with the target net $\bar{\rho}$.
- Loss over the $B = 1{,}024$ pairs: mean squared relative errors $(\text{target} - \text{prediction})/(\varepsilon_{\text{loss}} + |\text{target}|)$ on W's first value, its increments and λ; $\varepsilon_{\text{loss}}$: 1 to 0.01.
- Targets fixed; one Adam step on ρ at 10^{-6}; then $\bar{\rho} \leftarrow \tau\rho + (1-\tau)\bar{\rho}$, $\tau = 10^{-4}$: the target lags about $1/\tau = 10{,}000$ updates (Deck 4.1: 0.02, 50).
- **A supervised target from a slower copy, as Deck 4.1's critic; not a residual.**

Source: The paper, §6.3 (loss function) and §6.4 (Stage 3, hyperparameters).

The Training Loop in One Picture



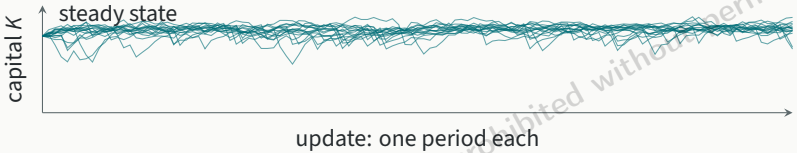
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- **1. Samples**: $E = 1{,}024$ pairs side by side; their current states are the dataset, $P = E$.
- **2. Update**: one Adam step on all $B = E = 1{,}024$ pairs: the mini-batch is the dataset.
- **3–4.** Nothing is reused: after the step every pair moves one period; the update is the period.
- **The look-ahead (n_z target calls, n_z clearings at $t + 1$) moves nothing.**

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Source: The paper, §6.4 (the outline and step 7). Symbols: this part's symbol page.

1. Samples: 1,024 Tracks, Moved One Period per Update



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- Start: Stage 1's steady state for all $E = 1{,}024$ pairs (the paper gives no initial histories); Stage 2 spreads them, continuation values held at the steady state's. 16 of E drawn.
- A pair's run never restarts: each update moves it one period (z' drawn, Γ' by the lottery, the history shifted): no pool, no branching, no episode. Dips: disasters.
- Coverage: histories drawn from Π^z meet the $3^{T_{\text{hist}}}$ windows at their true frequencies; a life is $J = 10$ periods, the window $T_{\text{hist}} = 100$; Γ comes along with its history.
- **Coverage is checked on simulated paths, not guaranteed.**

Source: The paper, §6.3 (Step 3) and §6.4 (Stages 1–2, step 7). Paths schematic.

2. Inside One Update, Line by Line

One Adam step on all $B = E = 1{,}024$ pairs $(z, \Gamma,$ history$)$

1. online network at each history: $\hat{W}$ and $\hat{\lambda}$ (one call)
2. target network at the $n_z = 3$ shifted histories (n_z calls)
3. clear labour at t: $n_{\text{Newton}} = 5$ steps, each an EGM solve of all J cohorts; prices, policies, λ
4. move Γ one period: n_z candidate successors
5. solve $t{+}1$ for each z' with the target network (n_z more clearings); the expectation: W^{tar}
6. loss; one Adam step at 10^{-6}; nudge the target network by τ
7. draw z' from Π^z; that successor is the pair's next state

- Per pair: $1 + n_z = 4$ network calls and $1 + n_z = 4$ clearings of $n_{\text{Newton}} = 5$ EGM solves each; times B: 4,096 calls and $(1 + n_z)\, n_{\text{Newton}} B = 20{,}480$ EGM solves an update.
- Three stages: (1) a steady state by EGM, which seeds the pairs; (2) aggregate risk on, continuation values held at the steady state, to spread them; (3) the loop above.

Source: The paper, §6.4 (the seven steps, Stages 1–3, hyperparameters).

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One Clock, Three Stages, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
network update	every period, on all $B = E = 1{,}024$ pairs	per pair: 4 network calls, 4 clearings of $n_{\text{Newton}} = 5$ EGM solves; one Adam step on ρ, the target nudged by τ
stage	three in all	none: 1 a steady state seeds the pairs; 2 aggregate risk on, continuation values held at the steady state's; 3 the loop

The algorithm: the paper gives no count of updates

1. Stage 1: a deterministic steady state by EGM; $E = 1{,}024$ pairs seeded from it
2. Stage 2: aggregate risk on, continuation values held at the steady state's: the pairs spread
3. Stage 3, repeat: one update on all $B = E$ pairs (previous page); move each pair one period

- **Unlike Part B's episode of $U_{\text{epis}} = 384$ updates on one cloud and Deck 4.2's pool of $P_{\max} = 128$: here the update is the period, and nothing is reused.**

Source: The paper, §6.4 (Stages 1–3, step 7).

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The Symbols

symbol	value	what it counts, and the formula
J, N_e, N_k, n_z	10, 3, 300, 3	cohorts of six years; income states; asset points; regimes
JN_eN_k	9,000	masses in $\Gamma(j, e, k)$; 6,000 with 200 points
T_{hist}	100	regimes in the window (the paper's T); at least a lifetime, J
n_{out}	8,101	continuation values for ages $1, \ldots, J-1$, $(J-1)N_eN_k$, plus $\hat{\lambda}$
n_{hid}	$3 \times 1{,}024$	hidden layers (GELU)
E, B	1,024, 1,024	pairs $(z, \Gamma,$ history$)$ side by side; the mini-batch is all of them
n_{Newton}	5	Newton steps to clear labour; each an EGM solve
n_{call}, n_{clear}	4, 4	per pair and update: network calls, $1 + n_z$; clearings, $1 + n_z$
n_{EGM}	20	EGM solves per pair and update, $(1 + n_z)\, n_{\text{Newton}}$; $\times B$: 20,480
rate, τ, $1/\tau$	10^{-6}, 10^{-4}, 10,000	Adam; the target's nudge; updates the target lags, about
$\varepsilon_{\text{loss}}$	1, 0.1, 0.01	the stabiliser in the relative errors, lowered twice
stages, U	3, not given	steady state, pretraining, the loop; updates in all: not stated

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- The paper's H, h, μ and T are J, j, Γ and T_{hist} here.

Source: The paper, §6.2 (parameters), §6.3 (implementation), §6.4 (hyperparameters).

Two More Pipelines, the Same Letters

	Deep Eq. Nets, KS	Deep Eq. Nets, OLG (B)	Sequence space, KS	Sequence space, OLG (C)
side by side	$E = 16$ tracks	$E = 8{,}192$ economies	$P_{\max} = 128$ pool	$E = 1{,}024$ pairs
a run starts from	its own end state	one seed, no restart	a parent from the pool	the steady state, no restart
periods per run	$T_{\text{epis}} = 128$	one	$T_{\text{gen}} = 100$	one
B of *P*	128 of 16,384	64 of *E*, in order	16 of ≤ 128	all *E*
per dataset	$U_{\text{epis}} = 128$, exhausted	$U_{\text{epis}} = 384$: 3 passes	$U_{\text{round}} = 10$, never exhausted	1, used once
datasets	65,000 episodes	10,340 episodes	10,000 rounds	every update
U in all	8,320,000	3,970,560	100,000	not given
look-ahead	one period, stored	one period, differentiated	one period, computed	one period, target net

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- **Same letters: the OLG versions differ in what a datum is and how long it lives.**

Source: Deck 4.2, *Three Pipelines, the Same Letters*; the two symbol pages of this deck.

Bridge: Each Tool for Its Question

Paths: one unanticipated shock, any distribution.

The state as input: when Γ is a short vector, as here.

The history as input: when Γ is many distributions.

Next: back to Deck 4.1's histogram, and a government that taxes on it.

Source: The sources of this deck.

- Age turns Γ into a stack. Without ε the stack is a vector of $2J$ numbers, and Deep Equilibrium Nets solve it exactly.
- With ε the stack is many histograms: the history is the input, continuation values the output, EGM the solver.
- **Deck 4.4: a government that cannot commit chooses a tax rule on the whole distribution.**

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